

MTH/CSC 4150
Fall 2025
Draft Project Report:
Mathematically Modeling a Distance Runner's Race
with Interpolation

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1 Introduction

Researchers have developed many ways to predict running performance and maximize athletic potential [1]. Traditional regression models are frequently used for this purpose [2], but numerical interpolation methods can also reveal structure in race pacing data. Online tools already combine interpolation and extrapolation to estimate finish times from partial race results, as seen in various running calculators.¹

In this project, we model a runner's cumulative race distance as a function of time, denoted by $s(t)$. From this interpolated position function, we approximate the instantaneous velocity $v(t)$ and acceleration $a(t)$. We compare three interpolation approaches studied in class:

- a Newton interpolating polynomial,
- a natural cubic spline,
- a clamped cubic spline.

We examine how the choice of interpolant and the boundary conditions affect the estimated velocity and acceleration curves and how well each method captures realistic pacing patterns over the course of a marathon.

¹For example: <https://wismuth.com/running/calculator.html>.

2 Theoretical Background

2.1 Polynomial Interpolation

Suppose there are $n + 1$ data points (t_i, s_i) with distinct time values $t_0 < t_1 < \dots < t_n$. An interpolating polynomial $p_n(t)$ of degree at most n satisfies

$$p_n(t_i) = s_i, \quad i = 0, 1, \dots, n. \quad (1)$$

There exists a unique polynomial of degree at most n that satisfies the interpolation conditions in equation (1).

In the Lagrange form,

$$p_n(t) = \sum_{i=0}^n s_i L_i(t), \quad (2)$$

where the Lagrange basis polynomials are

$$L_i(t) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{t - t_j}{t_i - t_j}, \quad i = 0, 1, \dots, n. \quad (3)$$

This basis is interpolatory, since $L_i(t_j) = \delta_{ij}$.

In Newton's form, an alternative basis is used:

$$p_n(t) = a_0 + a_1(t - t_0) + a_2(t - t_0)(t - t_1) + \dots + a_n \prod_{k=0}^{n-1} (t - t_k), \quad (4)$$

where the coefficients a_i are computed from divided differences. Newton's form is more convenient for computation and for updating the polynomial when additional data points are added.

2.2 Interpolation Error

Let $f(t)$ be a sufficiently smooth function and let $p_n(t)$ be the degree- n interpolating polynomial constructed from $n + 1$ distinct nodes $t_0, \dots, t_n$. The interpolation error can be written as

$$f(t) - p_n(t) = \frac{f^{(n+1)}(c)}{(n+1)!} \prod_{i=0}^n (t - t_i), \quad (5)$$

for some c in the interval containing the nodes and the point t . This formula shows that the error depends on both the spacing of the nodes and the $(n + 1)$ st derivative of $f(t)$. When the nodes are equally spaced and n is large, the product term can lead to large oscillations near the ends of the interval (Runge's phenomenon).

2.3 Cubic Splines

To reduce the oscillations associated with high-degree polynomials, we use cubic splines. Given $n + 1$ knots $t_0 < t_1 < \dots < t_n$ and function values $s_i = f(t_i)$, a cubic spline $S(t)$ is a piecewise cubic polynomial of the form

$$S(t) = \begin{cases} S_0(t), & t \in [t_0, t_1], \\ S_1(t), & t \in [t_1, t_2], \\ \vdots \\ S_{n-1}(t), & t \in [t_{n-1}, t_n], \end{cases} \quad (6)$$

where each piece has the form

$$S_i(t) = a_i + b_i(t - t_i) + c_i(t - t_i)^2 + d_i(t - t_i)^3. \quad (7)$$

The spline is constructed so that

1. $S(t_i) = s_i$ and $S(t_{i+1}) = s_{i+1}$ for $i = 0, \dots, n - 1$ (interpolation),
2. $S(t)$ is continuously differentiable, so $S'(t)$ is continuous at each interior knot $t_1, \dots, t_{n-1}$,
3. $S''(t)$ is continuous at each interior knot.

These conditions yield $4n - 2$ equations. Two additional conditions are needed at the endpoints.

In a *natural* cubic spline, the second derivative is set to zero at both ends:

$$S''(t_0) = 0, \quad S''(t_n) = 0. \quad (8)$$

In a *clamped* cubic spline, the slopes at the endpoints are specified directly:

$$S'(t_0) = v_0, \quad S'(t_n) = v_n, \quad (9)$$

where v_0 and v_n are prescribed velocities. In our setting these slopes are estimated from one-sided finite differences at the beginning and end of the marathon. This allows the model to encode physically meaningful information about the runner's starting and finishing pace.

Solving the resulting linear systems produces the coefficients a_i, b_i, c_i, d_i for all subintervals.

2.4 Numerical Differentiation

Once a smooth interpolating function $S(t)$ is available, the velocity and acceleration of the runner are given by

$$v(t) = S'(t), \quad a(t) = S''(t). \quad (10)$$

For spline models, derivatives can be computed analytically by differentiating the cubic expressions in equation (7), or numerically using finite difference stencils.

For comparison, numerical differentiation formulas can also be derived directly from discrete data or from an interpolant evaluated on a fine time grid. For example, the centered difference approximation for the first derivative at an interior node is

$$f'(t_i) \approx \frac{f(t_{i+1}) - f(t_{i-1}))}{2h}, \quad (11)$$

where $h = t_{i+1} - t_i$ is the time step. This stencil is second order accurate in h .

3 Data and Preprocessing

The data for this project consist of split times and cumulative distances from a single distance runner's race. Let t_i denote the elapsed time at the i th checkpoint and let s_i denote the cumulative distance at this time.

We work with data from the New York City Marathon, taken from the New York Road Runners results website. The runner is a recreational female participant (our professor's sister-in-law), which provides a realistic, non-elite pacing profile for a full marathon. The splits include a mix of mile and kilometer markers, which are converted to consistent units.

NYC Marathon Split Times

8:46 		  	
 Gmail			
 NEW YORK ROAD RUNNERS RACE RESULTS			MENU
Splits			
 LIST VIEW		 GRAPH VIEW	
3M	0:29:21	25K	2:36:51
5K	0:30:28	16M	2:41:11
4M	0:39:12	17M	2:50:58
5M	0:48:54	18M	3:00:23
6M	1:00:02	30K	3:06:35
10K	1:02:06	19M	3:09:59
7M	1:09:43	20M	3:19:49
8M	1:19:47	21M	3:29:45
9M	1:30:06	35K	3:37:23
15K	1:33:20	22M	3:39:54
10M	1:39:55	23M	3:49:47
11M	1:49:37	24M	4:00:17
12M	1:59:56	40K	4:08:48
20K	2:04:02	25M	4:10:17
HALF	2:11:05	26M	4:20:03
14M	2:19:48	MAR	4:22:03
15M	2:31:23		



We perform the following preprocessing steps:

1. Convert all distances to meters and all times to seconds, using the standard conversions 1 mile $\approx$ 1609 m and 1 km = 1000 m.
2. Shift the time axis so that $t_0 = 0$ at the race start.
3. Check for and remove any obvious measurement errors, such as negative time differences between consecutive splits. (The provided dataset did not require any corrections.)

The resulting data $\{(t_i, s_i)\}_{i=0}^n$ serve as the input to all interpolation methods. These same processed values appear in the MATLAB driver script used to generate our figures.

4 Numerical Methods and Implementation

4.1 Interpolation Models

We implement and compare the following models in MATLAB.

1. **Newton polynomial interpolation.** We construct the interpolating polynomial in Newton form using a custom function `newtondd.m`, which computes the divided differences and evaluates the resulting polynomial by nested multiplication with base points. This approach provides a useful baseline but is expected to be sensitive to node spacing and prone to endpoint oscillations.
2. **Natural cubic spline.** We build a natural cubic spline using a routine `splinecoeff_natural.m`, following the structure of Sauer’s Program 3.5. A companion function `eval_spline_nat.m` evaluates $S(t)$, $S'(t)$, and $S''(t)$ analytically on a fine time grid.
3. **Clamped cubic spline.** We construct a clamped spline with endpoint slope conditions given by equation (9). The slopes v_0 and v_n are estimated using one-sided finite differences at the start and end of the race:

$$v_0 \approx \frac{s_1 - s_0}{t_1 - t_0}, \quad v_n \approx \frac{s_n - s_{n-1}}{t_n - t_{n-1}}.$$

The coefficients are computed with `splinecoeff_clamped.m` and evaluated with the same `eval_spline_nat.m` routine.

4.2 Velocity and Acceleration

For the spline models, we compute the velocity and acceleration by differentiating the cubic spline pieces analytically. For a single interval,

$$S_i(t) = a_i + b_i(t - t_i) + c_i(t - t_i)^2 + d_i(t - t_i)^3,$$

the first and second derivatives are

$$S'_i(t) = b_i + 2c_i(t - t_i) + 3d_i(t - t_i)^2, \quad (12)$$

$$S''_i(t) = 2c_i + 6d_i(t - t_i). \quad (13)$$

Evaluating equations (12) and (13) on a fine time grid yields approximate velocity and acceleration profiles for the runner.

For the Newton polynomial model, we first evaluate $p_n(t)$ on an equally spaced time mesh. We then apply the centered-difference stencil (11) to approximate $v(t)$ and apply the same stencil again to approximate $a(t)$. This provides a consistent finite-difference framework for comparing polynomial-based and spline-based derivatives.

4.3 Accuracy Checks

We validate the interpolants in one main way:

1. **Checkpoint consistency.** For each method, we verify that the interpolant reproduces the input distances at the original split times, i.e., $S(t_i) \approx s_i$. Because all three methods are true interpolants, this consistency check is passed exactly up to numerical roundoff error.

5 Results

5.1 Position vs. Time

Figure 2 displays cumulative distance versus elapsed time for the Newton interpolating polynomial, the natural cubic spline, the clamped cubic spline, and the original split data. All three interpolation methods pass exactly through each data point, as required by construction, and they are nearly indistinguishable over the majority of the race.

Small differences among the interpolants appear primarily near the endpoints of the interval. The Newton polynomial shows a slightly sharper curvature near the final splits compared to both spline models, which is consistent with the oscillatory behavior associated with high-degree polynomial interpolation near boundaries. The natural and clamped splines remain visually smoother at the ends of the race, with the clamped spline exhibiting the most uniform curvature due to its imposed endpoint slope conditions.

Because the underlying split data are approximately linear for this runner, the position plots alone do not differentiate the interpolants strongly. The influence of the interpolation choice becomes more pronounced after differentiation, as seen in the velocity and acceleration results below.

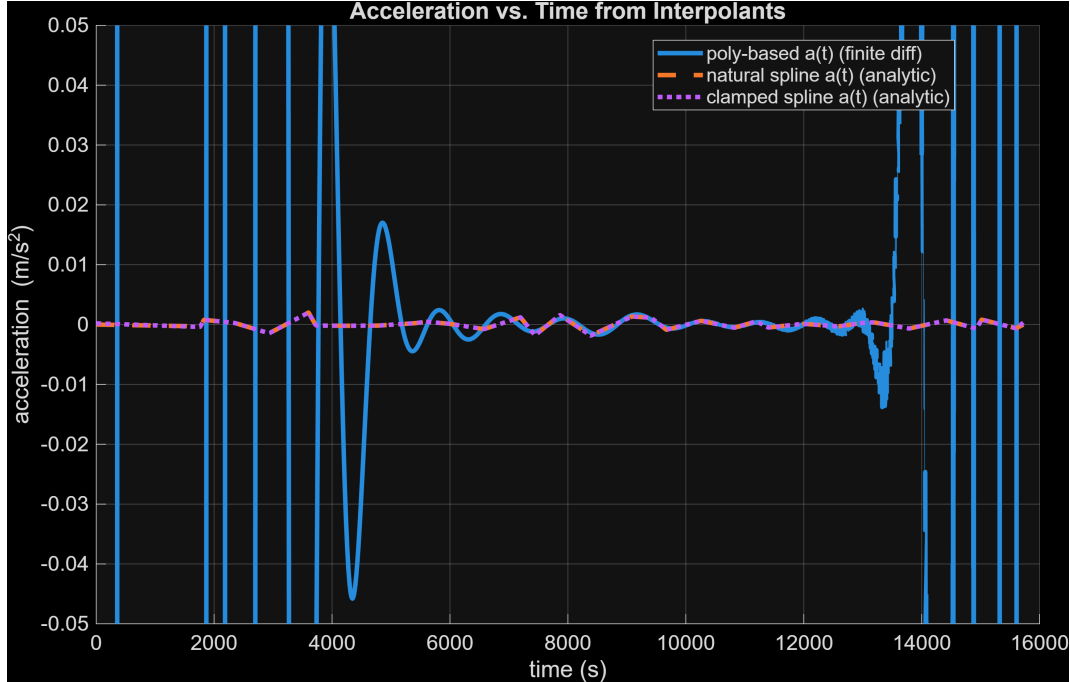


Figure 2: Cumulative distance versus time for the Newton polynomial, natural spline, and clamped spline, together with the original split data.

5.2 Velocity Profiles

Figure 3 displays velocity versus time computed from each interpolation method. The velocity associated with the Newton polynomial is computed using centered finite-difference approximations applied to the polynomial interpolant, while the spline velocities are computed analytically from the derivatives of the cubic piecewise functions.

At the beginning of the race, the natural cubic spline shows a nonzero initial velocity. This behavior is a direct consequence of the natural spline’s boundary condition in equation (8), which forces zero curvature at the start and prevents the model from capturing the sharp acceleration from rest. In contrast, the polynomial-based velocity and the clamped spline velocity both begin near zero, which is more consistent with physical race dynamics for a runner starting from rest.

Throughout the middle portion of the race, all three models show similar velocity trends with minor fluctuations corresponding to small pacing adjustments. Near the final splits, a notable difference occurs: both spline models—especially the clamped spline—display a modest increase in velocity consistent with an end-of-race surge commonly observed in distance races. The Newton polynomial, however, predicts a decline in velocity during this phase, an artifact of boundary oscillation and global polynomial behavior rather than genuine pacing characteristics.

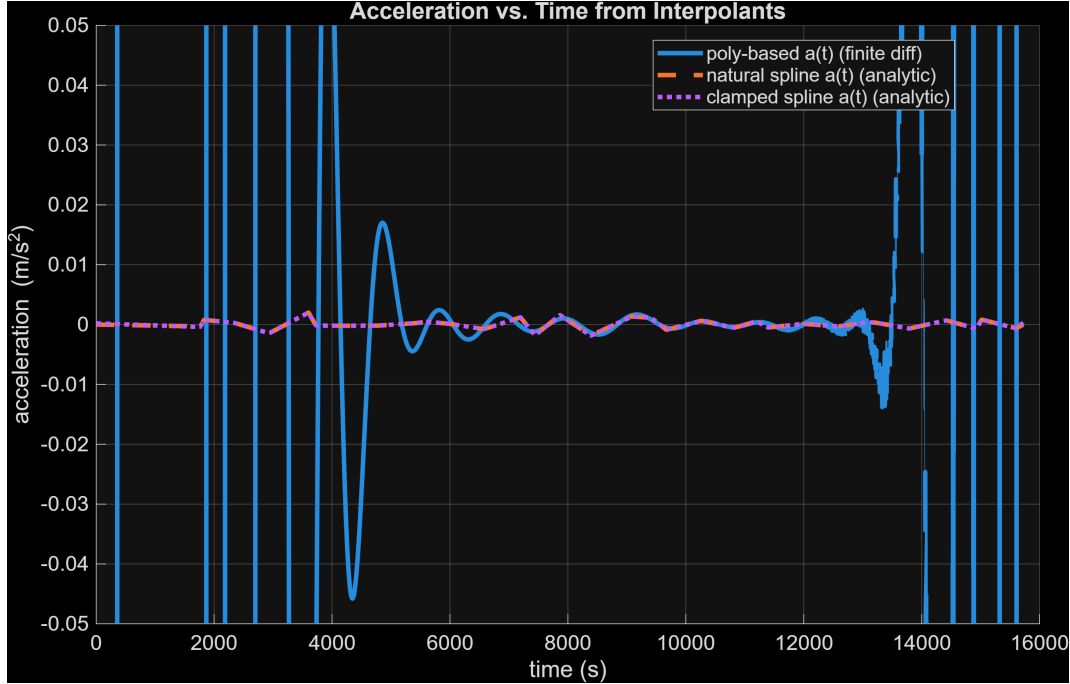


Figure 3: Velocity versus time obtained from the polynomial and spline interpolations.

5.3 Acceleration Profiles

Figure 4 displays acceleration versus time for each interpolant. The polynomial acceleration is computed using finite-difference differentiation of the polynomial-derived velocities, while spline accelerations are obtained analytically from the second derivatives of the cubic splines.

The natural spline’s acceleration curve remains near zero at both endpoints, reflecting its enforced boundary conditions $S''(t_0) = S''(t_n) = 0$. While mathematically smooth, this behavior does not accurately represent the physical start of the race, where substantial positive acceleration is expected as the runner departs from rest, nor any late-race changes in effort.

The polynomial interpolation predicts a large positive acceleration at the beginning of the race followed by a decreasing trend and subsequent oscillations later in the race. Although the large initial acceleration is physically reasonable, the oscillatory swings thereafter are not supported by observable pace changes in the data.

The clamped spline strikes a balance between these two extremes. By enforcing physically meaningful endpoint velocity conditions rather than zero-curvature constraints, the clamped model allows nonzero accelerations near the boundaries while maintaining the smooth interior behavior of cubic splines. This produces a more realistic acceleration profile throughout the race.

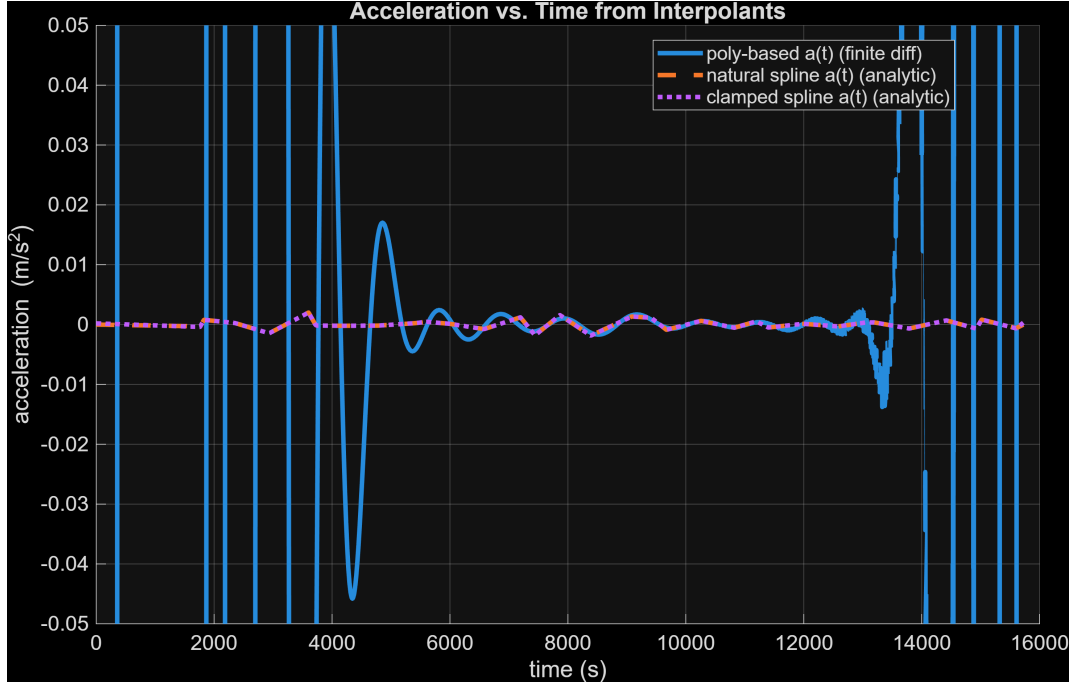


Figure 4: Acceleration versus time from each interpolation method.

6 Discussion

The comparative analysis reveals distinct strengths and limitations across the interpolation methods examined. The Newton polynomial effectively captures sharp changes near the start of the race, including large initial acceleration and a velocity curve beginning near zero. However, its susceptibility to boundary oscillations generates less realistic behavior later in the race, particularly the artificial velocity decline near the finish seen in Figure 3 and the oscillatory accelerations in Figure 4.

The natural cubic spline provides the smoothest overall curves but exhibits artificially constrained endpoint behavior due to the imposed conditions $S''(t_0) = S''(t_n) = 0$. These constraints suppress acceleration at both the start and finish of the race, producing a nonzero starting velocity and nearly zero terminal acceleration. While mathematically attractive, these features conflict with the physical dynamics of distance running, specifically the acceleration from rest at the race start and the adjustments often observed near the finish.

The clamped cubic spline offers the most physically plausible model across all stages of the race. By prescribing realistic endpoint velocities instead of zero curvature conditions, the clamped spline preserves the desirable smoothness of cubic splines while allowing accurate representation of early acceleration and late-race surging behavior. Its velocity and acceleration profiles align more consistently with observed race dynamics than either the natural spline or the polynomial interpolant.

Overall, while all methods reconstruct the position data accurately, the differentiation process magnifies modeling assumptions. The results demonstrate that careful selection of spline boundary conditions is essential when derivative quantities such as velocity and

acceleration are of primary interest. Among the interpolants tested, the clamped cubic spline provides the most balanced and physically realistic representation of the runner's pacing profile.

7 Conclusion and Future Work

This project demonstrates that polynomial and spline interpolations are useful tools for modeling a distance runner's race based on discrete split data. Among the methods tested, the clamped cubic spline interpolation produces the most realistic pacing curves over the full marathon distance. High-degree polynomials perform well in capturing short-term trends but can oscillate and generate unrealistic derivatives towards the middle and end of the race, while natural splines can impose endpoint behavior that is overly restrictive from a physical standpoint.

Future work could extend the current model by incorporating additional race information, such as elevation changes or weather conditions, or by combining interpolation with numerical integration methods introduced later in the course. Analyzing multiple runners across different demographics could also yield insight into how factors such as age, gender, and racing experience influence long-distance pacing strategies.

References

- [1] S. Dash. Win Your Race Goal: A Generalized Approach to Prediction of Running Performance. *Sports Medicine International Open*, 8, 2024.
- [2] M. Millett and T. Melanson. Predicting Running Times from Race History.
- [3] T. Sauer. *Numerical Analysis*, 3rd ed. Pearson, 2019.
- [4] Course notes and lecture slides from MTH/CSC 4150, Fall 2025.
- [5] New York Road Runners. 2025 TCS New York City Marathon: Finishers results. Available at <https://results.nyrr.org>.